

MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 10: Introduction to the Finite Volume Method, Control Volumes, and Godunov's Scheme

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1 Introduction: The Paradigm Shift from Nodes to Volumes

Throughout Lectures 6 through 9, we studied the **Finite Difference Method (FDM)** for hyperbolic equations. In Lecture 9, the Lax-Wendroff theorem revealed a critical requirement for computing discontinuous flows: discrete schemes must be written in **conservative flux form** to avoid generating artificial mass sinks and computing false shock speeds.

However, the Finite Difference framework suffers from an inherent conceptual contradiction when applied to compressible fluid flow:

- **Finite Difference is fundamentally differential:** It approximates pointwise partial derivatives $\frac{\partial u}{\partial t}$ and $\frac{\partial u}{\partial x}$ at discrete grid nodes x_j using Taylor series expansions.
- **Physical shocks are non-differential:** Across a shock wave or contact discontinuity, the physical state jumps discontinuously. Pointwise derivatives do not exist ($\frac{\partial u}{\partial x} \rightarrow \pm\infty$), rendering Taylor series expansions strictly invalid.

To compute discontinuous fluid flows robustly, we must abandon the pointwise differential approach and return to the **fundamental macroscopic physics**: *mass, momentum, and energy are conserved over finite regions of space, whether the underlying flow field is smooth or discontinuous.*

This insight is the foundation of the **Finite Volume Method (FVM)**. Instead of tracking point values $u(x_j, t)$ at grid vertices, FVM tracks **cell averages** over small spatial control volumes:

$$\bar{u}_i(t) \equiv \frac{1}{\Delta x_i} \int_{x_{i-1/2}}^{x_{i+1/2}} u(x, t) dx \quad (1)$$

Because the integral of a discontinuous, bounded function is perfectly well-behaved, cell averages remain smooth and physically meaningful even when sharp shock discontinuities traverse the computational grid!

In this lecture, we begin **Module 3: Finite Volume Methods**:

1. We formulate the exact integral conservation law over space-time control volumes.
2. We demonstrate that the discrete Finite Volume balance equation is **mathematically exact**—all numerical approximation is localized to the interface fluxes.
3. We solve the local **Riemann problem** at cell interfaces for both linear advection and the nonlinear Burgers' equation.
4. We develop **Godunov's pioneering 1959 method**, the foundation of modern shock-capturing CFD.
5. We examine the mathematical properties of Godunov's scheme: monotonicity, the TVD property, entropy satisfaction, and first-order numerical diffusion.

2 The 1D Control Volume Formulation

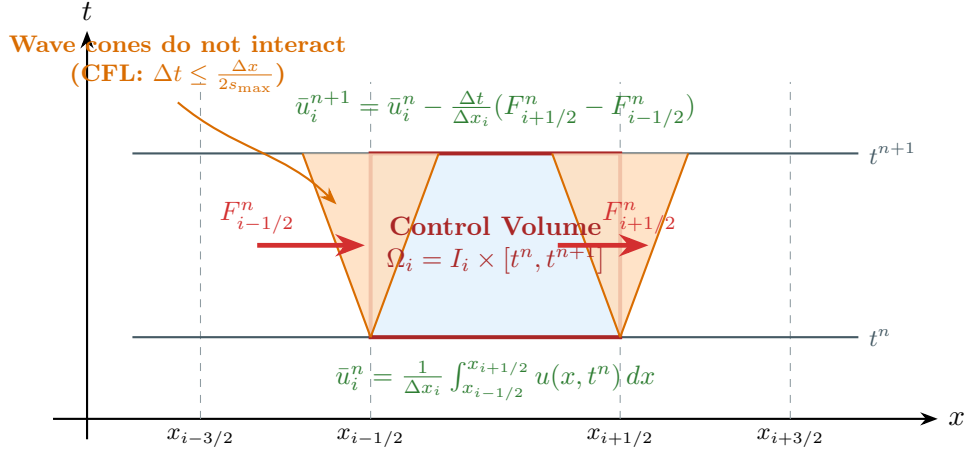


Figure 1: Space-time control volume $\Omega_i = [x_{i-1/2}, x_{i+1/2}] \times [t^n, t^{n+1}]$. Cell averages $\bar{u}_i^n$ and $\bar{u}_i^{n+1}$ represent the state integrated along the bottom and top boundaries. Changes in cell average are driven exclusively by the net time-integrated fluxes $F_{i-1/2}^n$ and $F_{i+1/2}^n$ crossing the left and right cell interfaces.

2.1 Space-Time Integration of the Conservation Law

Consider the scalar one-dimensional hyperbolic conservation law:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \quad (2)$$

Divide the spatial domain into disjoint computational cells (control volumes) $I_i = [x_{i-1/2}, x_{i+1/2}]$ of width $\Delta x_i = x_{i+1/2} - x_{i-1/2}$, with cell centers $x_i = \frac{1}{2}(x_{i-1/2} + x_{i+1/2})$. The time domain is discretized into time intervals $[t^n, t^{n+1}]$ with step size $\Delta t = t^{n+1} - t^n$.

Now integrate equation (2) over the two-dimensional **space-time control volume**:

$$\begin{aligned} \Omega_i^n &\equiv [x_{i-1/2}, x_{i+1/2}] \times [t^n, t^{n+1}] \\ \iint_{\Omega_i^n} \left(\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} \right) dx dt &= 0 \end{aligned} \quad (3)$$

Splitting the double integral into spatial and temporal components:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} \left(\int_{t^n}^{t^{n+1}} \frac{\partial u}{\partial t} dt \right) dx + \int_{t^n}^{t^{n+1}} \left(\int_{x_{i-1/2}}^{x_{i+1/2}} \frac{\partial f(u)}{\partial x} dx \right) dt = 0 \quad (4)$$

Applying the Fundamental Theorem of Calculus to both inner integrals:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} [u(x, t^{n+1}) - u(x, t^n)] dx + \int_{t^n}^{t^{n+1}} [f(u(x_{i+1/2}, t)) - f(u(x_{i-1/2}, t))] dt = 0 \quad (5)$$

Rearranging terms so that the state at the new time level t^{n+1} is isolated:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} u(x, t^{n+1}) dx = \int_{x_{i-1/2}}^{x_{i+1/2}} u(x, t^n) dx - \left[\int_{t^n}^{t^{n+1}} f(u(x_{i+1/2}, t)) dt - \int_{t^n}^{t^{n+1}} f(u(x_{i-1/2}, t)) dt \right] \quad (6)$$

2.2 The Exact Finite Volume Update Equation

Using the definition of the cell average $\bar{u}_i$ from equation (1), we define the **exact time-averaged numerical flux** through cell interface $x_{i+1/2}$ over the time interval $[t^n, t^{n+1}]$:

$$F_{i+1/2}^n \equiv \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(u(x_{i+1/2}, t)) dt \quad (7)$$

Dividing equation (6) by the cell length Δx_i yields the foundational update formula of the Finite Volume Method:

$$\bar{u}_i^{n+1} = \bar{u}_i^n - \frac{\Delta t}{\Delta x_i} (F_{i+1/2}^n - F_{i-1/2}^n) \quad (8)$$

The Central Miracle of Finite Volume Methods

Equation (8) is **mathematically exact**. No Taylor series truncation, spatial differencing, or temporal discretization errors have been introduced!

If one could evaluate the exact time-averaged boundary fluxes $F_{i\pm 1/2}^n$, equation (8) would yield the *exact* cell averages of the true physical solution for all time.

Therefore, in the Finite Volume framework, **all numerical modeling is concentrated into a single task**: approximating the interface flux $F_{i+1/2}^n$ given only the discrete cell averages $\{\bar{u}_i^n\}$.

2.3 Discrete Global Conservation

Summing equation (8) multiplied by Δx_i over all cells in a domain spanning $i = 1, 2, \dots, N$:

$$\sum_{i=1}^N \bar{u}_i^{n+1} \Delta x_i = \sum_{i=1}^N \bar{u}_i^n \Delta x_i - \Delta t \sum_{i=1}^N (F_{i+1/2}^n - F_{i-1/2}^n) \quad (9)$$

Because the numerical flux leaving cell i through its right face ($F_{i+1/2}^n$) is identically equal to the flux entering cell $i+1$ through its left face, the interior summation telescopes to zero:

$$\sum_{i=1}^N (F_{i+1/2}^n - F_{i-1/2}^n) = F_{N+1/2}^n - F_{1/2}^n \quad (10)$$

Hence, the total discrete mass inside the domain changes exclusively through the physical boundaries:

$$\sum_{i=1}^N \bar{u}_i^{n+1} \Delta x_i = \sum_{i=1}^N \bar{u}_i^n \Delta x_i - \Delta t (F_{N+1/2}^n - F_{1/2}^n) \quad (11)$$

Exact conservation is guaranteed on any grid (uniform or non-uniform) and for any flow (smooth or containing shocks)!

3 The Interface Dilemma and the Riemann Problem

3.1 The Dilemma at the Interface

To evaluate the flux integral (7), we must know the state $u(x_{i+1/2}, t)$ along the vertical interface line $x = x_{i+1/2}$ for $t \in [t^n, t^{n+1}]$.

However, the computational data at time t^n consists only of cell averages $\{\bar{u}_i^n\}$. If we represent the flow inside each cell by its cell average (piecewise constant):

$$u(x, t^n) = \bar{u}_i^n \quad \text{for } x \in [x_{i-1/2}, x_{i+1/2}] \quad (12)$$

then at the interface $x_{i+1/2}$, the state is **discontinuous**:

$$u(x_{i+1/2}^-, t^n) = \bar{u}_i^n \equiv u_L, \quad u(x_{i+1/2}^+, t^n) = \bar{u}_{i+1}^n \equiv u_R \quad (13)$$

What is the state $u(x_{i+1/2}, t)$ for $t > t^n$? Does the left state u_L prevail, or the right state u_R , or does an entirely new intermediate state form?

3.2 Definition of the Riemann Problem

The mathematical problem of determining how an isolated jump discontinuity evolves in time under a conservation law is called the **Riemann Problem** (first studied by Bernhard Riemann in 1860 for gas dynamics):

The Riemann Problem for a Scalar Conservation Law

The Riemann problem is the initial value problem (IVP) for a conservation law subject to piecewise-constant initial data with a single jump:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0, \quad u(x, 0) = \begin{cases} u_L, & x < 0 \\ u_R, & x > 0 \end{cases} \quad (14)$$

Because the differential equation and the initial data contain no intrinsic length scale or time scale, the solution is **self-similar**: it depends only on the similarity ratio $\xi = x/t$:

$$u(x, t) = W\left(\frac{x}{t}\right) \quad (15)$$

Any feature (shock or characteristic) emanates from the origin ($x = 0, t = 0$) along a straight ray in the (x, t) plane.

3.3 Riemann Solution for Linear Advection

For linear advection $f(u) = au$ ($a \in \mathbb{R}$):

$$u_t + au_x = 0 \quad (16)$$

The jump simply propagates along the characteristic ray $x = at$ without changing shape:

$$u(x, t) = u_0(x - at) = \begin{cases} u_L, & x - at < 0 \iff \frac{x}{t} < a \\ u_R, & x - at > 0 \iff \frac{x}{t} > a \end{cases} \quad (17)$$

Along the interface ray $x = 0$, the state is:

$$u^*(x = 0, t > 0) = \begin{cases} u_L, & \text{if } a > 0 \\ u_R, & \text{if } a < 0 \end{cases} \quad (18)$$

This confirms that the physical upwind state is the exact solution of the linear Riemann problem along the cell boundary!

3.4 Riemann Solution for Convex Scalar Conservation Laws (Burgers' Equation)

For a nonlinear conservation law with strictly convex flux ($f''(u) > 0$, e.g., Burgers' equation $f(u) = \frac{1}{2}u^2$), the solution depends on the relative ordering of u_L and u_R .

3.4.1 Case 1: Shock Wave ($u_L > u_R$)

When $u_L > u_R$, characteristic lines from the left (speed $f'(u_L)$) travel faster than characteristic lines from the right (speed $f'(u_R)$) and collide. The unique weak solution satisfying the Oleinik entropy condition is a **discontinuous shock wave** propagating at speed s given by the Rankine-Hugoniot condition:

$$s = \frac{f(u_L) - f(u_R)}{u_L - u_R} \quad (19)$$

For Burgers' equation, $s = \frac{1}{2}(u_L + u_R)$. The solution is:

$$u(x, t) = \begin{cases} u_L, & x < st \\ u_R, & x > st \end{cases} \quad (20)$$

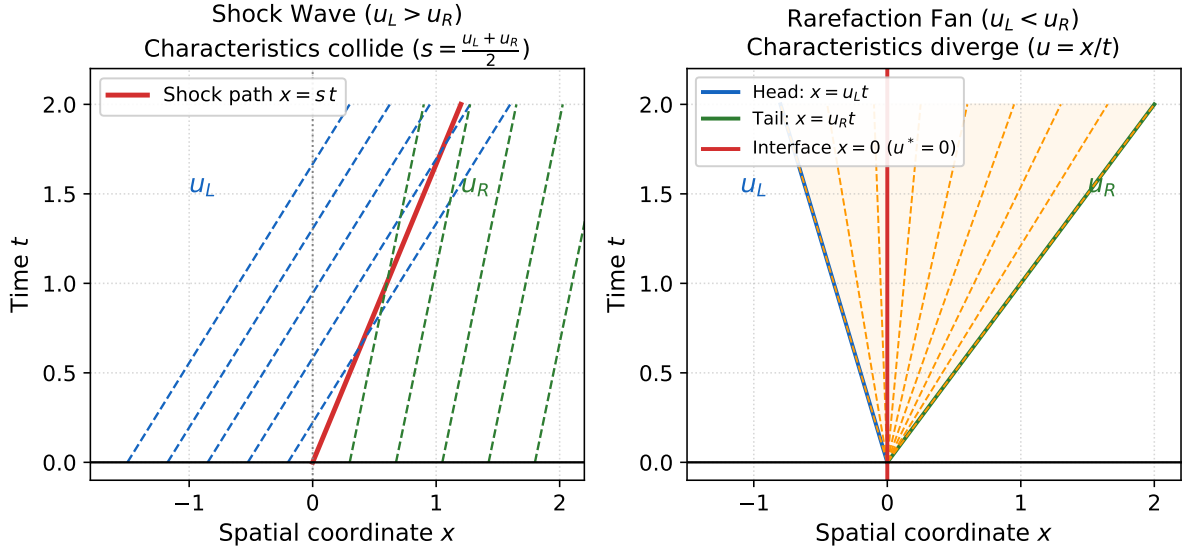


Figure 2: Structure of the Riemann problem in the (x, t) plane for a convex flux ($f''(u) > 0$). Left: Shock wave ($u_L > u_R$), characteristics converge into a shock moving at speed s . Right: Rarefaction fan ($u_L < u_R$), characteristics diverge and create a continuous expansion fan bounded by $x = f'(u_L)t$ and $x = f'(u_R)t$.

Along the interface $x = 0$, the state is:

$$u^*(0) = \begin{cases} u_L, & \text{if } s > 0 \\ u_R, & \text{if } s < 0 \end{cases} \quad (21)$$

3.4.2 Case 2: Rarefaction Wave ($u_L < u_R$)

When $u_L < u_R$, characteristics diverge and leave an empty region between $x = f'(u_L)t$ and $x = f'(u_R)t$. The entropy-satisfying weak solution is a smooth, continuous **rarefaction fan** (expansion fan).

Substituting the similarity ansatz $u(x, t) = W(\xi)$ with $\xi = x/t$ into equation (2):

$$\frac{\partial u}{\partial t} + f'(u) \frac{\partial u}{\partial x} = -\frac{\xi}{t} W'(\xi) + \frac{1}{t} f'(W(\xi)) W'(\xi) = \frac{W'(\xi)}{t} [f'(W(\xi)) - \xi] = 0 \quad (22)$$

Inside the fan, $W'(\xi) \neq 0$, which yields the characteristic speed equation:

$$f'(W(\xi)) = \xi \implies W(\xi) = (f')^{-1}(\xi) \quad (23)$$

For Burgers' equation, $f'(u) = u$, so the inverse function is simply $(f')^{-1}(\xi) = \xi = x/t$. Thus, the complete rarefaction solution is:

$$u(x, t) = \begin{cases} u_L, & \frac{x}{t} \leq u_L \\ \frac{x}{t}, & u_L < \frac{x}{t} < u_R \\ u_R, & \frac{x}{t} \geq u_R \end{cases} \quad (24)$$

3.4.3 The Transonic Rarefaction and the Sonic Point

When $u_L < 0 < u_R$, the expansion fan straddles the interface line $x/t = 0$. The interface state is evaluated directly by setting $\xi = 0$ in equation (24):

$$u^*(0) = 0 \quad (\text{the sonic point where } f'(u) = 0) \quad (25)$$

The flux at the interface is therefore $f(u^*(0)) = f(0) = 0$. This is the famous **transonic rarefaction**: the flow expands from subsonic (or left-moving) to supersonic (or right-moving) across the stationary cell interface!

4 Godunov's Foundational Method (1959)

In 1959, the Soviet mathematician Sergei K. Godunov proposed a breakthrough methodology: *use the exact solution of local Riemann problems to compute the numerical fluxes of a finite volume scheme.*

4.1 The Three Steps of Godunov's Scheme

Given the cell averages $\{\bar{u}_i^n\}$ at time level t^n :

Step 1. Reconstruction: Construct a piecewise-constant function $\tilde{u}(x, t^n)$ by setting the state in each cell I_i equal to its cell average:

$$\tilde{u}(x, t^n) = \bar{u}_i^n \quad \text{for } x \in [x_{i-1/2}, x_{i+1/2}] \quad (26)$$

Step 2. Evolution: Solve the exact Riemann problem at each cell interface $x_{i+1/2}$ with left state $u_L = \bar{u}_i^n$ and right state $u_R = \bar{u}_{i+1}^n$. This generates waves propagating outward from each interface.

Step 3. Cell Averaging (Flux Evaluation): Under the **CFL condition**:

$$\Delta t \leq \frac{\Delta x}{2 \max |f'(u)|} \quad (27)$$

waves from interface $x_{i-1/2}$ and interface $x_{i+1/2}$ do not interact within the time interval Δt . Therefore, along the vertical interface line $x = x_{i+1/2}$, the state remains **constant in time**:

$$u(x_{i+1/2}, t) = u_{\text{RP}}(0; \bar{u}_i^n, \bar{u}_{i+1}^n) \equiv u_{i+1/2}^* \quad \text{for all } t \in [t^n, t^{n+1}] \quad (28)$$

Because the interface state $u_{i+1/2}^*$ is constant over the entire time step, the exact time-integral in the flux definition (7) simplifies trivially:

$$F_{i+1/2}^{\text{Godunov}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(u_{i+1/2}^*) dt = f(u_{i+1/2}^*) \quad (29)$$

Definition of the Godunov Numerical Flux

The **Godunov numerical flux** is the physical flux evaluated at the self-similar interface state of the exact Riemann problem:

$$F_{i+1/2}^{\text{Godunov}} = f(u_{\text{RP}}(0; \bar{u}_i^n, \bar{u}_{i+1}^n)) \quad (30)$$

4.2 General Closed-Form Godunov Flux for Convex Fluxes

For any scalar conservation law with strictly convex flux ($f''(u) > 0$), Osher (1984) showed that Godunov's flux can be written compactly as:

$$F^{\text{Godunov}}(u_L, u_R) = \begin{cases} \min_{u \in [u_L, u_R]} f(u), & \text{if } u_L \leq u_R \quad (\text{rarefaction}) \\ \max_{u \in [u_R, u_L]} f(u), & \text{if } u_L > u_R \quad (\text{shock}) \end{cases} \quad (31)$$

4.3 Explicit Formulas for Common Conservation Laws

4.3.1 Linear Advection ($f(u) = au$)

For $f(u) = au$, the min/max formula (31) yields:

$$F^{\text{Godunov}}(u_L, u_R) = \begin{cases} au_L, & a > 0 \\ au_R, & a < 0 \end{cases} = \frac{a}{2}(u_L + u_R) - \frac{|a|}{2}(u_R - u_L) \quad (32)$$

Substituting this flux into the FVM update (8) recovers the **classical first-order upwind scheme**! This proves that upwinding is not an ad-hoc trick—it is the exact Finite Volume discretization of linear wave propagation!

4.3.2 Inviscid Burgers' Equation ($f(u) = \frac{1}{2}u^2$)

For $f(u) = \frac{1}{2}u^2$, the sonic point occurs at $u_s = 0$ where $f'(0) = 0$. Evaluating formula (31) leads to the explicit five-branch decision rule:

$$F^{\text{Godunov}}(u_L, u_R) = \begin{cases} \frac{1}{2}u_L^2, & u_L \geq u_R \text{ and } s = \frac{u_L+u_R}{2} > 0 \quad (\text{right-moving shock}) \\ \frac{1}{2}u_R^2, & u_L \geq u_R \text{ and } s = \frac{u_L+u_R}{2} \leq 0 \quad (\text{left-moving shock}) \\ \frac{1}{2}u_L^2, & u_L < u_R \text{ and } u_L \geq 0 \quad (\text{supersonic right rarefaction}) \\ \frac{1}{2}u_R^2, & u_L < u_R \text{ and } u_R \leq 0 \quad (\text{supersonic left rarefaction}) \\ 0, & u_L < 0 < u_R \quad (\text{transonic rarefaction: sonic point } u^* = 0) \end{cases} \quad (33)$$

5 Mathematical Properties of Godunov's Scheme

5.1 Monotonicity and Godunov's Scheme

A finite difference or finite volume scheme expressed in the operator form:

$$\bar{u}_i^{n+1} = \mathcal{H}(\bar{u}_{i-1}^n, \bar{u}_i^n, \bar{u}_{i+1}^n) \quad (34)$$

is defined to be **monotone** if $\mathcal{H}$ is a non-decreasing function of each of its arguments:

$$\boxed{\frac{\partial \mathcal{H}}{\partial \bar{u}_{i-1}} \geq 0, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_i} \geq 0, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_{i+1}} \geq 0} \quad (35)$$

Theorem: Monotonicity of Godunov's Scheme

Under the standard CFL condition:

$$\nu \equiv \max |f'(u)| \frac{\Delta t}{\Delta x} \leq 1$$

Godunov's scheme is a **monotone scheme**.

Proof Sketch for Linear Advection. For $f(u) = au$ with $a > 0$, the Godunov scheme is:

$$\bar{u}_i^{n+1} = \mathcal{H}(\bar{u}_{i-1}^n, \bar{u}_i^n) = \bar{u}_i^n - \nu(\bar{u}_i^n - \bar{u}_{i-1}^n) = \nu \bar{u}_{i-1}^n + (1 - \nu) \bar{u}_i^n$$

Taking partial derivatives:

$$\frac{\partial \mathcal{H}}{\partial \bar{u}_{i-1}} = \nu \geq 0, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_i} = 1 - \nu \geq 0 \iff \nu \leq 1, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_{i+1}} = 0 \geq 0$$

All partial derivatives are non-negative if and only if $\nu \leq 1$. □

5.2 The TVD (Total Variation Diminishing) Property

The **Total Variation** of a discrete grid function $\bar{\mathbf{u}}$ is defined as:

$$\text{TV}(\bar{\mathbf{u}}) \equiv \sum_{i=-\infty}^{\infty} |\bar{u}_{i+1} - \bar{u}_i| \quad (36)$$

A scheme is **Total Variation Diminishing (TVD)** if:

$$\text{TV}(\bar{\mathbf{u}}^{n+1}) \leq \text{TV}(\bar{\mathbf{u}}^n) \quad (37)$$

A fundamental theorem of Harten (1983) establishes:

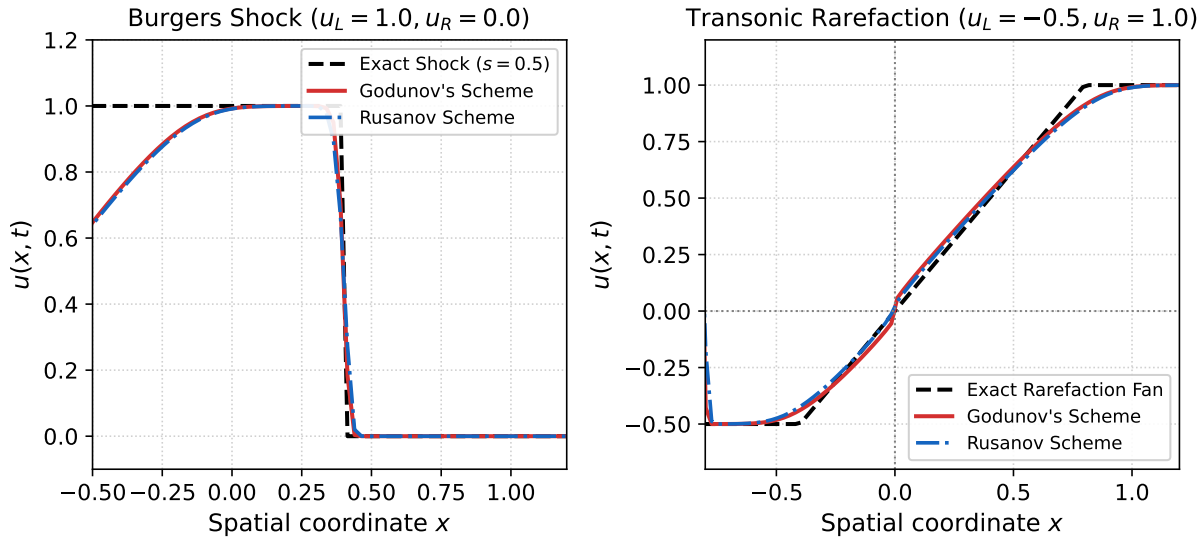


Figure 3: Numerical simulations with Godunov's scheme and Rusanov's scheme. Left: Moving Burgers shock ($u_L = 1.0, u_R = 0.0$); Godunov resolves the shock crisply and monotonically. Right: Transonic rarefaction ($u_L = -0.5, u_R = 1.0$); Godunov resolves the smooth expansion fan through the sonic point $u = 0$ without entropy violation.

Every monotone scheme is Total Variation Diminishing (TVD).

Consequences of the TVD property:

- **No new local extrema:** The scheme cannot create spurious wiggles, ripples, or overshoots near discontinuities ($u_{\min}^n \leq \bar{u}_i^{n+1} \leq u_{\max}^n$).
- **Monotonicity preservation:** If the initial data $\bar{\mathbf{u}}^0$ is monotonically decreasing (such as a shock profile), the numerical solution remains monotonically decreasing for all time steps $n > 0$.

5.3 Entropy Satisfaction

Because Godunov's method samples the *exact* Riemann solution, and the exact Riemann solution uniquely selects the entropy-satisfying wave structure (viscous shock instead of expansion shock), **Godunov's scheme automatically satisfies the discrete entropy condition**. Unphysical expansion shocks never appear!

5.4 Order of Accuracy and First-Order Numerical Diffusion

Why does Godunov's scheme not violate Godunov's Barrier Theorem (Lecture 8)? Because **Godunov's scheme is strictly first-order accurate** ($\mathcal{O}(\Delta x, \Delta t)$).

The first-order accuracy stems directly from **Step 1 (Piecewise Constant Reconstruction)**. By discarding sub-cell gradient information and averaging the flow into flat piecewise blocks, the scheme introduces a projection error of order $\mathcal{O}(\Delta x)$.

Taylor series expansion of the Godunov update reveals its modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \underbrace{\frac{a \Delta x}{2} (1 - \nu)}_{\nu_{\text{num}}} \frac{\partial^2 u}{\partial x^2} + \mathcal{O}(\Delta x^2) \quad (38)$$

The numerical diffusion coefficient $\nu_{\text{num}} > 0$ smears sharp discontinuities over several grid cells and rounds off smooth peaks. To achieve high-order accuracy while preserving monotonicity, we must use **high-order reconstruction (MUSCL / slope limiters)**, which we will develop in Lecture 11.

6 Approximate Riemann Solvers: The Rusanov (Local Lax-Friedrichs) Flux

Solving the exact nonlinear Riemann problem at every cell interface is straightforward for scalar equations, but for systems like the 3D compressible Euler equations, solving the exact iterative Riemann problem at every interface in a mesh of millions of cells is computationally prohibitive!

Modern CFD relies extensively on **Approximate Riemann Solvers** that capture the interface flux efficiently. The simplest and most universal approximate flux is the **Rusanov flux** (also called the **Local Lax-Friedrichs flux**):

$$F_{i+1/2}^{\text{Rusanov}} = \frac{1}{2} (f(\bar{u}_i) + f(\bar{u}_{i+1})) - \frac{1}{2} s_{\max} (\bar{u}_{i+1} - \bar{u}_i) \quad (39)$$

where $s_{\max} = \max(|f'(\bar{u}_i)|, |f'(\bar{u}_{i+1})|)$ is the maximum local wave speed.

6.1 Physical Structure of the Rusanov Flux

The Rusanov flux decomposes into two intuitive components:

1. **Central Average:** $\frac{1}{2}(f(\bar{u}_i) + f(\bar{u}_{i+1}))$, which is second-order accurate in space but unconditionally unstable by itself.
2. **Scalar Dissipation:** $-\frac{1}{2}s_{\max}(\bar{u}_{i+1} - \bar{u}_i)$, which acts as an artificial viscosity proportional to the jump across the interface, stabilizing the scheme.

As shown in Figure 3, Rusanov is more dissipative than Godunov, smearing shocks over 4–5 cells instead of 2–3 cells, but it requires zero case-by-case logic and extends effortlessly to systems of equations!

7 Summary: Finite Difference vs. Finite Volume

Feature	Finite Difference Method (FDM)	Finite Volume Method (FVM)
Primary Unknown	Point value $u(x_j, t)$ at grid nodes	Cell average $\bar{u}_i(t) = \frac{1}{\Delta x} \int_{I_i} u \, dx$
Governing Equation	Differential form: $u_t + f_x = 0$	Integral form over each cell I_i
Behavior at Shocks	Taylor series breaks; derivatives blow up	Cell averages remain bounded and well-behaved
Discrete Conservation	Requires careful flux differencing	Guaranteed by construction via telescoping fluxes
Mesh Flexibility	Structured Cartesian or mapped grids	Arbitrary unstructured meshes (polygons, polyhedra)
Flux Evaluation	Differences of nodal point values	Exact or approximate Riemann solvers

8 Exercises

Exercises for Lecture 10

E1. Godunov Flux for Linear Advection with Negative Velocity: Let $u_t + au_x = 0$ with constant speed $a < 0$.

- Solve the Riemann problem at interface $x = 0$ for $t > 0$.
- Evaluate the exact interface state $u^*(0)$ and show that $F^{\text{Godunov}}(u_L, u_R) = au_R$.
- Substitute this flux into the FVM update formula (8) and show that it yields the forward-differenced upwind scheme.
- Determine the exact range of the CFL number $\nu = \frac{|a|\Delta t}{\Delta x}$ for which this scheme is monotone.

E2. Transonic Rarefaction Calculation: Consider Burgers' equation $u_t + (\frac{1}{2}u^2)_x = 0$ with cell averages $\bar{u}_1 = -2.0$ and $\bar{u}_2 = +3.0$ across interface $x_{3/2}$.

- Is the Riemann solution a shock or a rarefaction fan? Justify your answer using characteristic speeds.
- Determine the propagation speeds of the left edge and the right edge of the wave.
- Find the exact state $u^*(0)$ along the interface line $x/t = 0$.
- Compute the Godunov flux $F_{3/2}^{\text{Godunov}}$ and compare it to the Rusanov flux $F_{3/2}^{\text{Rusanov}}$. Explain why the Rusanov flux is non-zero while the Godunov flux is identically zero.

E3. Monotonicity of the Rusanov Scheme for Scalar Conservation Laws: Consider the general scalar conservation law $u_t + f(u)_x = 0$ discretized using the Finite Volume Method with the Rusanov flux (39).

- Write the full update operator $\bar{u}_i^{n+1} = \mathcal{H}(\bar{u}_{i-1}^n, \bar{u}_i^n, \bar{u}_{i+1}^n)$ assuming a constant maximum wave speed $s_{\max} = \alpha > 0$.
- Compute the partial derivatives $\frac{\partial \mathcal{H}}{\partial \bar{u}_{i-1}}$, $\frac{\partial \mathcal{H}}{\partial \bar{u}_i}$, and $\frac{\partial \mathcal{H}}{\partial \bar{u}_{i+1}}$.
- Prove that the scheme is monotone provided that the CFL condition $\alpha \frac{\Delta t}{\Delta x} \leq 1$ is satisfied.

References

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